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Polysaccharide of *Hericium erinaceus* attenuates colitis in C57BL/6 mice via regulation of oxidative stress, inflammation-related signaling pathways and modulating the composition of the gut microbiota

Yilin Ren[a,b,c], Yan Geng[b,*], Yan Du[b], Wang Li[b], Zhen-Ming Lu[a,c], Hong-Yu Xu[a,c], Guo-Hua Xu[d], Jin-Song Shi[b], Zheng-Hong Xu[a,b,c,**]

[a]*School of Biotechnology, Jiangnan University, Wuxi 214122, PR China*
[b]*School of Pharmaceutical Science, Jiangnan University, Wuxi 214122, PR China*
[c]*National Engineering Laboratory for Cereal Fermentation Technology, Jiangnan University, Wuxi 214122, PR China*
[d]*Jiangsu Shenhua Pharmaceutical Co., Ltd., 188 Shenhua Road, Huai'an, Jiangsu 211600, PR China*

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Abstract

Inflammatory bowel disease (IBD) is a disease caused by a dysregulated immune with unknown etiology. *Hericium erinaceus* (*H. erinaceus*) is a Chinese medicinal fungus, with the effect of prevention and treatment of gastrointestinal disorders. In this study, we have tested the anti-inflammatory effect of polysaccharide of *H. erinaceus* (HECP, Mw: 86.67 kDa) in the model of dextran sulfate sodium (DSS)-induced colitis in C57BL/6 mice. Our data indicated that HECP could improve clinical symptoms and down-regulate key markers of oxidative stresses, including nitric oxide (NO), malondialdehyde (MDA), total superoxide dismutase (T-SOD), and myeloperoxidase (MPO). HECP also suppressed the secretion of interleukin (IL)-6, interleukin (IL)-1β, tumor necrosis factor (TNF)-α and the expression of cyclooxygenase-2 (COX-2), inducible nitric oxide synthase (iNOS) and decreased the expression of related mRNA. Meanwhile, HECP blocked phosphorylation of nuclear factor-κB (NF-κB) p65, NF-κB inhibitor alpha (IκB-α), mitogen-activated protein kinases (MAPK) and Protein kinase B (Akt) in DSS-treated mice. Moreover, HECP reversed DSS-induced gut dysbiosis and maintained intestinal barrier integrity. In conclusion, HECP ameliorates DSS-induced intestinal injury in mice, which suggests that HECP can serve as a protective dietary nutrient against IBD.
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Keywords: Inflammatory bowel disease; Anti-inflammatory; Gut dysbiosis; *Akkermansia*; *PICRUSt*

1. Introduction

Inflammatory bowel disease (IBD) that includes ulcerative colitis (UC) and Crohn's disease (CD) is characterized by chronic inflammation of the intestinal tract [1]. The pathogenesis of IBD remains unclear, but inflammation and oxidative stress are considered as the essential mechanism underlying the pathophysiology of IBD [2].

Nuclear factor-κB (NF-κB), mitogen-activated protein kinases (MAPK) and phosphatidylinositol-3-kinase (P3IK)/protein kinase B (Akt) pathways are major inflammatory signaling pathways, which regulate the expression of inflammatory cytokines and proteins [3,4]. Cytokines, as essential mediators between activated immune and non-immune cells, are involved in intestinal immune inflammation of IBD [1]. Inflammation augments oxidative stress by stimulating reactive oxygen/nitrogen species (ROS/RNS)-generating systems, such as myeloperoxidase (MPO) and inducible nitric oxide synthase (iNOS), and oxidative stress is intimately involved in inflammatory cytokines and infiltration of inflammatory cells via stimulating NF-κB signaling pathway in IBD [5,6]. Nitric oxide (NO) production catalyzed by iNOS and expression of cyclooxygenase-2 (COX-2) catalyzed the production of prostaglandins can reflect the degree of inflammation [7]. The gut microbiota is part of a complex network responsible for maintaining normal physiological function. Recent studies indicate that gut microbiota are associated with the development of IBD [8,9]. Intestinal dysbacteriosis could cause IBD and reduce the microbial diversity in mice [10,11]. In genetically susceptible individuals, the gut mucosal integrity is impaired, and the microbial antigens escape through the

Abbreviations: Akt, Protein kinase B; CD, Crohn's disease; COX-2, cyclooxygenase-2; DAI, disease activity index; DSS, dextran sulfate sodium; ERK, extracellular signal-regulated kinase; HECP, Polysaccharide of *Hericium erinaceus* (Mw: 86.67 kDa); *H. erinaceus*, *Hericium erinaceus*; IBD, inflammatory bowel disease; IL, interleukin; iNOS, inducible nitric oxide synthase; JNK, c-Jun N-terminal kinase; MAPK, mitogen-activated protein kinases; MDA, malondialdehyde; MPO, myeloperoxidase; NF-κB, nuclear factor-κB; NO, nitric oxide; PICRUSt, Phylogenetic Investigation of Communities by Reconstruction of Unobserved States; PI3K, phosphatidylinositol-3-kinase; qRT-PCR, quantitative real-time polymerase chain reaction; ROS/RNS, reactive oxygen/nitrogen species; T-SOD, total superoxide dismutase; UC, ulcerative colitis; 5-ASA, 5-aminosalicylic acid; SCFAs, short chain fatty acids; TNF-α, tumor necrosis factor α.

* Corresponding author.
E-mail addresses: gengyan@jiangnan.edu.cn (Y. Geng), zhenghxu@jiangnan.edu.cn (Z.-H. Xu).

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epithelial barrier more easily, thereby activating inappropriate immune response or underlying chronic inflammation [12]. Previous studies have shown that prebiotics are non-digestible, fermentable carbohydrates and fibers, such as polysaccharides, which might be useful in preventing and treating acute and chronic conditions including antibiotic associated diarrhea and IBD [11,13]. Intestinal bacteria possesses distinct polysaccharide preferences. On the other hand, polysaccharides may favor the growth of specific bacterial species and modulate the composition of the gut microbiota [11,13–15]. Also, prebiotics improve the intestinal tight junction integrity and decrease blood endotoxemia caused by LPS [13,16].

With edible values and medicinal properties, *Hericium erinaceus* (*H. erinaceus*) has a potential therapeutic effect on anticancer, immune stimulation, improving lipid metabolism and gastrointestinal diseases. Aqueous extract and polysaccharide of *H. erinaceus* can reduce ulceration, provide cytoprotection in ethanol-induced gastric ulcers to protect against gastric mucosal damage [17–22]. A protein isolated from *H. erinaceus* could improve the immune system via regulation the composition and metabolism of gut microbiota [23]. Our previous study showed that the ethanol extract from *H. erinaceus* had anti-inflammatory effects, which contributed to the effective amelioration of DSS-induced colitis symptoms in mice [24]. However, very little is known about the role of polysaccharide from *H. erinaceus* on colitis. The aim of this study was to investigate the potential protective effect of the polysaccharide of fermented *H. erinaceus* mycelium powder (HECP) against DSS-induced colitis in C57BL/6 mice.

2. Materials and methods

2.1. Materials

The dry power of fermented *H. erinaceus* mycelium was obtained from Jiangsu Shenhua Pharmaceutical Co., Ltd., China., and prepared as Fig. S1. Dextran sulfate sodium (DSS, 36–50kD) was purchased from MP Biomedical (Solon, OH, USA). 5-ASA (5-aminosalicylic acid) was purchased from Tokyo Chemical Industry Co., Ltd, Japan. Trizol, SYBR Green PCR Master Mix and BCA Protein Assay Kit were obtained from Thermo Fisher Scientific (MA. USA). MPO, NO, total superoxide dismutase (T-SOD) and malondialdehyde (MDA) assay kits were purchased from Jiancheng Bioengineering Institute (Nanjing, China). Enzyme linked immunosorbent (ELISA) assay kits for TNF-α, IL-1β and IL-6 were obtained from eBioscience (San Diego, CA, USA). Antibodies for p-p65, p65, p-IκBα, IκBα, p-Akt, Akt, p-p38, p38, p-ERK, ERK, p-JNK, JNK and GAPDH were purchased from Cell Signaling Technology (Danvers, MA, USA). Antibodies for iNOS and COX-2 were purchased from Abcam (Cambridge, UK). Polyvinylidene difluoride (PVDF) membranes were purchased from GE Healthcare (England, UK). Unless otherwise specified, all chemicals were purchased from Sangon Biotech (Shanghai, China).

2.2. Chemical analysis of HECP

Total carbohydrate content was determined by the phenol sulfuric acid method. Reducing sugars were measured by the dinitrosalicylate method. Glucose was used for the standard curve. Total protein content was determined by the Lowry assay with BSA for the standard curve. The homogeneity and molecular weight of HECP were determined by high performance gel permeation chromatography (HPGPC), which was performed on a 6000 high performance liquid chromatograph (Waters, MA, USA). Ultrahydrogel™ Linear (300 mm×7.8 mmid×2) was maintained at 45 °C and the mobile phase was 0.1 M $NaNO_3$ (flow rate=0.9 ml/min) and detected by a Waters 2414 refractive index detector. The sample (2 mg) was dissolved in the mobile phase and centrifuged. A 20 μl sample was injected in each run. Pullulan polysaccharide calibration kit (Agilent technologies, CA, USA) of Mw 180, 4600, 21,400, 41,100, 133,800 and 200,000 Da was used to calibrate the average molecular weights. All samples were filtered through a 0.22 μm membrane before injection. The monosaccharide composition of HECP was determined by gas chromatography (GC) analysis. The tested sample (20 mg) dissolved in 2 ml 1 M H_2SO_4, was hydrolyzed at 100 °C for 4 h, and converted to their alditol acetates as previously described and were analyzed by GC. GC was performed on an Agilent 6280 instrument fitted with FID and equipped with an OV1701 fused-silica capillary (30 m×0.53 mm×1.00 μm). The column temperature was increased from 120 to 240 °C at a rate of 3 °C /min then hold on 10 min (Tables S1–S3).

2.3. Animals and treatment

Six-week-old male C57BL/6 mice of clean grade [SCXK(Hu)2012–002] were purchased from Shanghai SLAC Laboratories Animal Co. Ltd.. Six mice were housed per cage under a 12 h light/dark cycle at 22 °C for 7 days before the experiment. All mouse procedures and protocols were approved by the Institutional Animal Care and Use Committee of Jiangnan University, Wuxi, China [Approval No. JN. No20150103–20,150,119(2)]. Acute colitis was induced through oral administration of 2% DSS w/v in fresh tap water ad libitum for 7 days. Animals were randomly assigned to five groups: Control (CTL) group (mice received water), DSS group (mice received 2% DSS added in water from day 0), 5-ASA group (mice received 150 mg/kg/d 5-ASA by gavage and 2% DSS from day 0), 250 mg/kg HECP group (mice received 250 mg/kg/d HECP by gavage and 2% DSS from day 0), 500 mg/kg HECP group (mice received 500 mg/kg/d HECP by gavage and 2% DSS from day 0). Mice weight was recorded daily and animals were sacrificed at day 8. Colons were removed and submitted to macroscopic examination. The disease activity index (DAI), a score used to assess the severity of colitis, was evaluated as previously described [25]. Colon samples were rinsed with cold PBS, blotted dry, weighed, and their lengths were measured. Distal colon parts were fixed in 4% p-formaldehyde in PBS, embedded in paraffin and cut into sections of 4 μm. Then, hematoxylin and eosin (H&E) staining was used to observe the morphological changes under a light microscope [26].

2.4. Measurement of NO in serum

The blood samples were centrifuged (3000 rpm, 4 °C, 15 min) and the serum was collected. NO content in serum was measured using NO assay kit as the manufacturer's protocol.

2.5. Assay of MPO activity, MDA, T-SOD and inflammatory cytokines in colon tissue

The colonic sample was homogenized in ice-cold 50 mmol/L PBS solution (1:9, w/v). The activity of MPO was measured according to the manufacturer's instructions by MPO assay kit. The result was expressed as activity units per mg tissue. T-SOD and MDA were detected by assay kits as the manufacturer's protocols. The protein content was assayed by BCA protein assay kit and values were normalized to tissue protein levels. IL-1β, IL-6 and TNF-α cytokines were quantified using ELISA kits according to manufacturer's instruction. The results were expressed as per ml supernatant.

2.6. Analyses of relative mRNA expression in colon tissue

Total RNA was isolated from colonic tissues using the Trizol reagent (Thermo Fisher Scientific, MA. USA). Quantitative real-time polymerase chain reaction (qRT-PCR) was performed with a Real-time PCR Sequence Detection System (Bio-Rad Laboratories, CA, USA) using SYBR Green PCR Master Mix according to the manufacturer's protocol (Thermo Fisher Scientific, MA. USA). Primer sequences of IL-6, IL-1β, TNF-α, iNOS, COX-2 and GAPDH were shown in Table S4. The relative amount of each transcript was normalized to the amount of GAPDH in the same cDNA. All samples were run in duplicate in a single 96-well reaction plate, and data were analyzed according to the $2^{-\Delta\Delta Ct}$ method.

2.7. Western blot analysis

All colons were homogenized to extract total protein. Protein contents were quantified with the BCA protein assay kit (Thermo, USA); 30 μg protein was separated by 10% sodium dodecyl sulfate-polyacrylamide gel electrophoresis and transferred onto PVDF membranes. The membranes were blocked with 5% w/v defatted milk diluted in TBS-0.1% v/v Tween 20 for 1 h and incubated overnight at 4 °C with primary antibodies. The membranes were washed and incubated with secondary antibodies at room temperature for 2 h. Finally, the immunoreactive bands were visualized with an enhanced chemiluminiscence system (Bio-Rad Laboratories, CA, USA) and quantified using Image J software.

2.8. Bioinformatic analysis

Genomic DNA was extracted from every stool sample using the QIAamp DNA Stool Mini Kit according to manufacturer's protocols (QIAGEN, USA). Extracted DNA was diluted to 100 ng/μL, and mixed each group of samples together. The primers 338F (5′-barcode-ACTCCTACGGGAGGCAGCA-3′) and 806R (5′- GGACTACHVGGGTWTCT AAT-3′), where barcode is an eight-base sequence unique to each sample, were used to amplify the V3-V4 region of the bacteria 16S ribosomal RNA gene. Sequencing was performed using an Illumina (La Jolla, California) MiSeq. Pair-end reads generated by 16S sequencing were merged by using the QIIME v 1.9.1 script join_paired_ends.py and the fastq-join method to obtain~460 bp V3-V4 16S sequences. Merged reads were analyzed with QIIME v1.9.1. First, reads with N bases were removed. Reads containing three or more consecutive low-quality bases (Q<20) were truncated and the length of a trimmed read should be≥75% of its original length. Reads passing the quality filter were aligned to the Greengenes Database (version Aug, 2013) for chimera check by USEARCH v6.1. Then, UCLUST4 was applied for OTU clustering at the 97% similarity level. Ribosomal Database Project (RDP) classifier v2.25 was retrained with the Greengenes Database and utilized to assign a taxonomic rank to each representative OTU. OTU relative abundances were hierarchically clustered and visualized by the R software (v3.0.1) gplots package (heatmap.2 function). The heatmap was generated for scaled, normalized titer data using a Euclidean distance function with complete linkage clustering. Phylogenetic Investigation of Communities by Reconstruction of

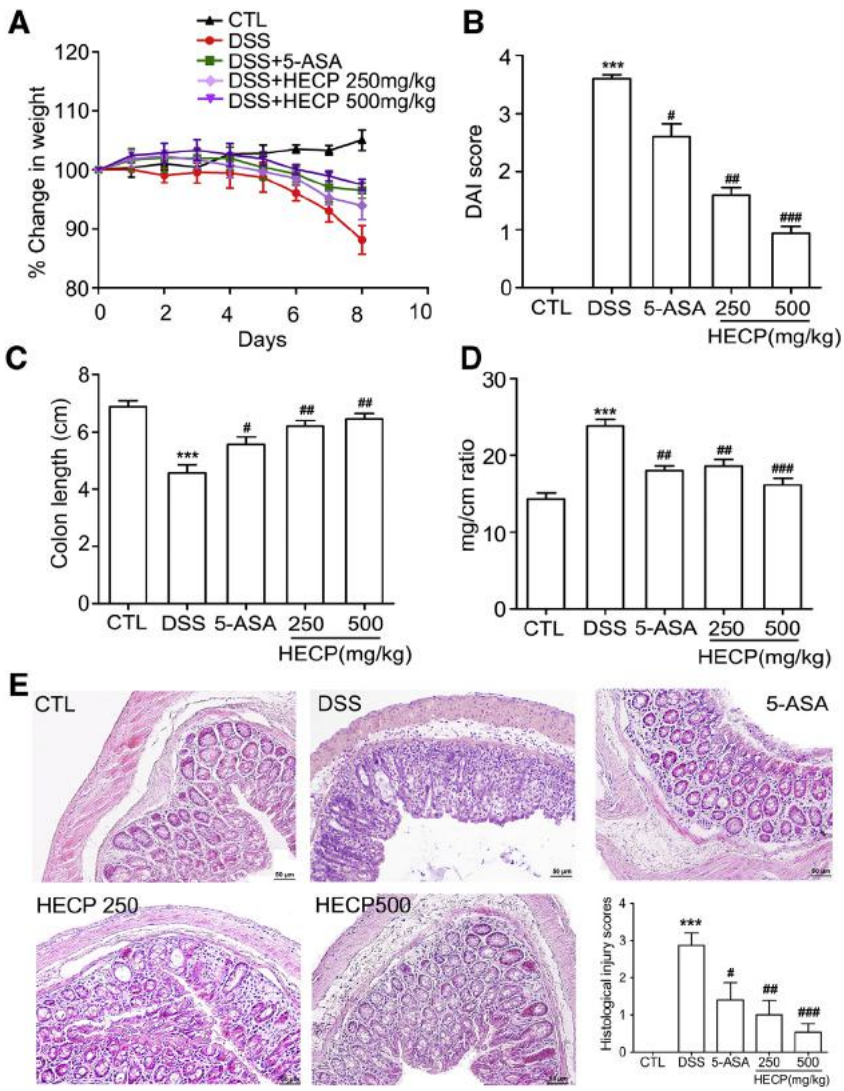

Fig. 1. Effect of HECP on clinical symptoms of DSS-induced colitis in C57BL/6mice. Mice were orally administered daily with HECP for 7 days during DSS administration. Mice were sacrificed at day 8 and colons were harvested and submitted to macroscopic examination. $n=6$. (A) Body weight change (%) of mice were measured daily during the experiment; (B) Disease activity index (DAI) score after seven days were measured by body weight loss, rectal bleeding, and diarrhea; (C) Colon length; (D) Colon mg/cm ratio from each group of mice; (E) H&E staining of representative colons from untreated mice or mice treated with DSS (2%) and HECP or 5-ASA are shown under the same microscope (200×). Average histological injury scores (mean±SD; $n=6$) of each group. $^{*}P<.05$, $^{**}P<.01$, and $^{***}P<.001$ vs. the CTL group. $^{\#}P<.05$, $^{\#\#}P<.01$ and $^{\#\#\#}P<.001$ vs. the DSS group.

Unobserved States (PICRUSt) (http://picrust.github.com) software was used to predicted functions of KEGG categories represented in mice after HECP treatment.

2.9. Statistical analysis

The significance of differences between experimental groups was determined by using One-way analysis of variance (ANOVA) with post hoc Tukey test using the GraphPad Prism software (version 5.00, GraphPad Software, San Diego, California, USA), and if the $P<.05$, results were considered statistically significant.

3. Results

3.1. HECP ameliorated clinical symptoms in mice treated with DSS

Mice that only received 2% DSS daily didn't have large float about their body weight during the first five days, but showed a noticeable body weight loss after five days (Fig. 1A). During the experimental time, many clinical signs including diarrhea and rectal bleeding were happened gradually, and became worsen, as indicated in DAI score [25] (Fig. 1B). Mice from CTL group did not lose weight markedly and show any clinical symptoms during seven days, giving the value of 0 in DAI score. Treatment with HECP or positive control 5-ASA ameliorated body weight loss and reduced the presence of diarrhea and blood in feces (Fig. 1A and B).

Colitis induced by DSS is characterized as colon edema and shortened colon length, so colon length can be an indicator to reflect the severity of inflammation. We observed a notable shortening of the colon and an increment of mg/cm ratio in mice that had received DSS in comparison to mice of the control group (Fig. 1C and D). However, this condition was ameliorated in HECP group.

From the histopathology, administration of DSS had induced a chronic inflammation, characterized by colonic mucosal ulcers, epithelium disruption, and a significant immune cell infiltration, but treatment with 5-ASA or HECP significantly reduced the histological severity (Fig. 1E).

3.2. HECP improved oxidative stress and inhibited MPO activity in DSS-induced colitis

To assess whether HECP could affect oxidative stress in colon, we investigated NO concentration, MDA content, and T-SOD activity.

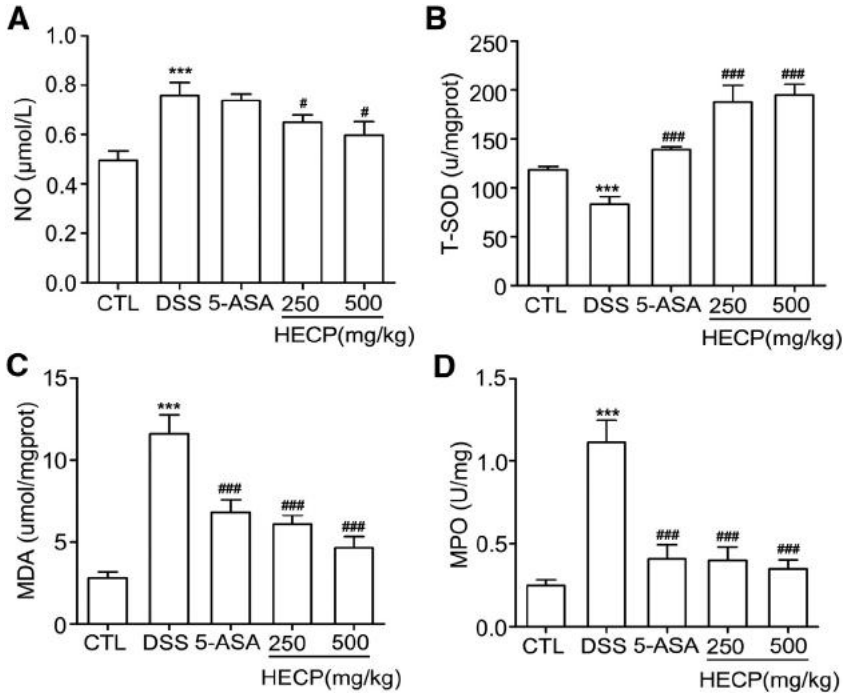

Fig. 2. Effect of HECP on oxidative damage of mucosa in DSS-induced colitis in C57BL/6 mice. (A) NO production; (B) T-SOD activity changes; (C) MDA production; (D) MPO activity. $^{*}P<.05$, $^{**}P<.01$, and $^{***}P<.001$ vs. the CTL group. $^{\#}P<.05$, $^{\#\#}P<.01$ and $^{\#\#\#}P<.001$ vs. the DSS group.

Compared to the normal group, the concentration of NO and MDA increased, while T-SOD activity had significantly decreased in colonic tissue of DSS group (Fig. 2A-C). These indicators were improved in HECP group (Fig. 2A-C). As the shown in Fig. 2D, MPO activity increased more than four fold in the colon of the DSS group compared with CTL group. Compared to the DSS group, 5-ASA and HECP group all suppressed MPO accumulation in colonic tissue with significant differences (Fig. 2D). These data indicated that 5-ASA and HECP both had a significantly effect on relieving the severity of colitis.

3.3. HECP suppressed production of inflammatory cytokines and proteins in colon tissue with DSS-induced colitis

Inflammatory cytokines IL-1β, TNF-α and IL-6 levels in colon tissue increased markedly in DSS-treated mice compared with the CTL group. Treatment with HECP or 5-ASA significantly suppressed IL-1β, TNF-α and IL-6 levels in colitis mice (Fig. 3A-C). As expected, DSS treatment induced the gene expression of key inflammatory cytokines (IL-1β, TNF-α, and IL-6) in the colon of colitis mice, while the administration of HECP could ameliorate that effect (Fig. 3D). In DSS-induced colitis, the colonic tissue exhibited higher protein levels of COX-2 and iNOS. Treatment with HECP significantly decreased DSS induced COX-2 and iNOS. Similar results were observed in mice treated with 5-ASA (Fig. 3E).

3.4. HECP blocked signaling pathways related to DSS-induced colitis

Activation of NF-κB, PI3K/AKT, and MAPK leads to the gene expression of inflammatory cytokines and other mediators involved in the pathogenesis of IBD [4]. As shown in Fig. 4A, we detected a significant increase in the phosphorylation of NF-κB p65 and IκBα levels in DSS group compared with the CTL group, suggesting the activation of NF-κB signaling pathway. After treatment with HECP, there was a marked reduction of NF-κB protein phosphorylation levels in the colon tissue. Phosphorylated Akt level increased in colon samples from DSS group, while treatment with HECP inhibited activation of this pathway by reducing Akt phosphorylation (Fig. 4B). The administration of DSS also significantly increased the phosphorylation of MAPK (P38, ERK, and JNK) in DSS group compared with the control group, while HECP alleviated the increased phosphorylation of MAPK induced by DSS (Fig. 4C). Thus, we deduced that HECP could attenuate DSS-induced colitis in mice through blocking phosphorylation of NF-κB, Akt, and MAPK.

3.5. HECP regulates structural segregation of gut microbiota in mice

To examine whether HECP regulated gut microbiota, we performed 16S rDNA sequencing in fecal samples. As shown in Fig. S3, the Simpson and Shannon indices revealed that the microbial diversity in CTL and HECP groups was higher than in the DSS group. Histograms illustrating the gut microbiota community structure revealed the microbial species and their relative abundance (Fig. S4). Among them, the most abundant phylums were Bacteroidetes, Firmicutes, and Verrucomicrobia. DSS treatment up regulated Verrucomicrobia and Actinobacteria, meanwhile down regulated Bacteroidetes in fecal samples compared with CTL group. HECP could reverse these changes (Fig. S4). HECP could partly reverse some varieties of DSS enhanced bacterial species, including *Arthrobacter* spp., *Methylibium* sp., *Succinivibrio* sp. and *Akkermansia muciniphila* and decreased *Desulfovibrio* sp (Fig. 5). Notably, HECP could partly reverse these changes. In some OTUs, 250 mg/kg HECP group was better than 500 mg/kg HECP group (OTU0593043, OTU0315147, OTU0007433, and OTU1110312). Collectively, these results showed that HECP modulated the gut microbiota of DSS-treated mice, resulting in a microbiota composition similar to that of CTL mice (Figs. 6, S4 and Supplementary Data 1).

As shown in Fig. 6, we used PICRUSt to predict functions of KEGG categories represented in mice after HECP treatment. The PICRUSt outputs show 256 differentially abundant KEGG pathways of the CTL and DSS Group, 55 of them are the metabolic pathways (54 increased and 1 decreased), including those microbial gene functions related to the glycan biosynthesis metabolism, amino acid metabolism, energy metabolism, lipid metabolism, glycan biosynthesis and metabolism,

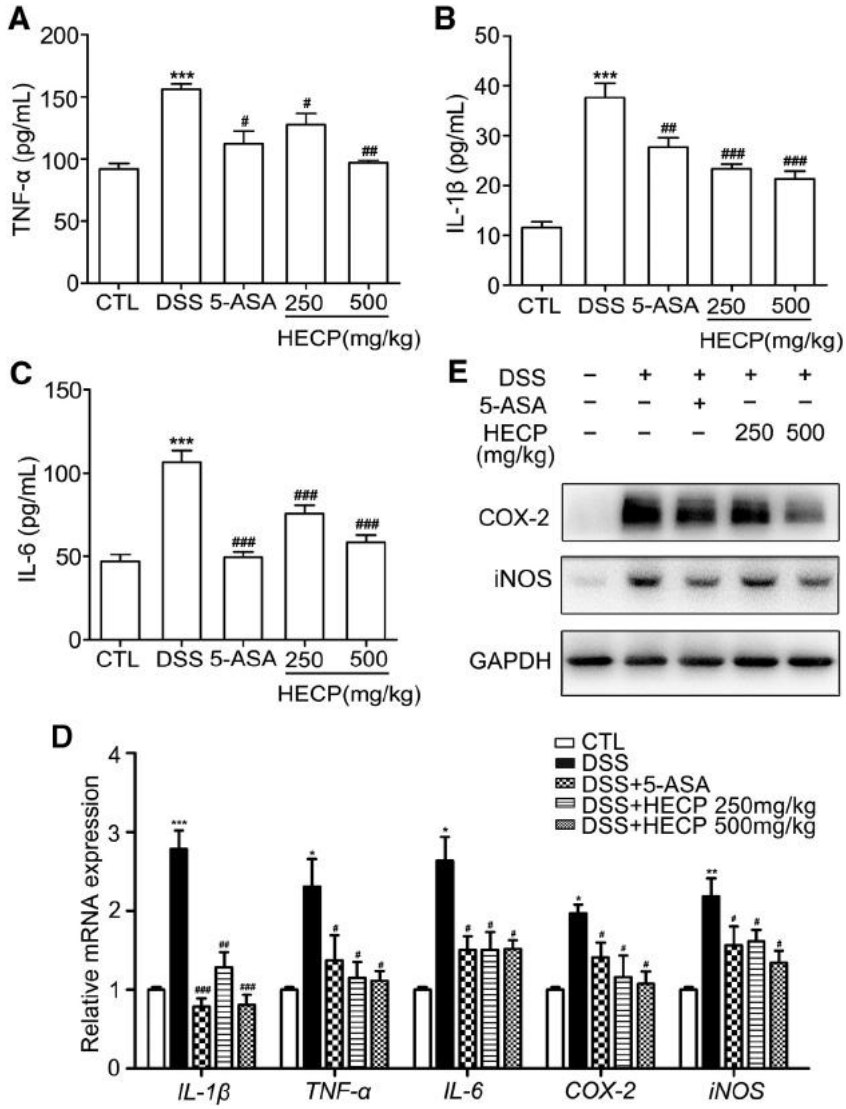

Fig. 3. Effect of HECP on inflammatory cytokines and proteins in colon tissue from DSS-induced colitis in C57BL/6 mice. (A) IL-1β, (B) TNF-α and (C) IL-6 levels in colonic homogenates were determined by ELISA at concentration of pg/mL; (D) relative expression of genes detected by qRT-PCR; (E) expression of inflammatory proteins detected by Western blot were determined by densitometry and normalized to GAPDH content for each sample. *$P<.05$, **$P<.01$, and ***$P<.001$ vs. the CTL group. #$P<.05$, ##$P<.01$ and ###$P<.001$ vs. the DSS group.

carbohydrate metabolism. These changes may be related to metabolic disorders and inflammatory reactions. Compared with DSS group, 500 mg/kg HECP treatment reversed the abundance of these metabolic pathways. Notably, the amino acids metabolism pathways (red) and the glycan biosynthesis pathways (green) of 500 mg/kg HECP group was reduced, ranging between 39.85%–69.70%, as well as 250 mg/kg HECP group.

4. Discussion

Colitis induced by DSS in C57BL/6 mice is characterized by the colon inflammation accompanied with diarrhea, rectal bleeding, body weight loss and shortening of the colorectum and an increased mucosal infiltration of inflammatory cells [27]. Current research on managing inflammation has focused on the use of natural products to offer alternative and effective anti-inflammatory therapies with low toxicity and minimal side effects [28]. Many researches have demonstrated that polysaccharides, naturally occurring substances derived from plants or mushroom, exhibit favorable therapeutic and health-promoting benefits, particularly in relation to diseases associated with inflammation [29–31]. *H. erinaceus* is a medicinal and edible mushroom and has been confirmed as a useful treatment for digestive diseases including gastric ulcer, chronic atrophic gastritis, and colitis [21,22,24]. Based on the protective effects of *H. erinaceus* on the gastrointestinal tract, we undertook this study to examine the effect of HECP on DSS-induced colitis in C57BL/6 mice via regulating oxidative stress, inhibiting expression of pro-inflammatory cytokines and acting on signaling pathways of NF-κB, AKT, and MAPK. 5-ASA at 150 mg/kg is the reference drug, since it demonstrated a clear preventive effect on DSS-induced colitis model in mice in previous studies carried out by Qing et al. [24]. Because both DSS and HECP are oral, we cannot exclude the possibility that HECP interacts with DSS leading to less colitogenecity of DSS.

Targeted oxidative stress and inflammation have become a therapeutic strategy for IBD. Oxidative stress has been proposed as a mechanism underlying the pathophysiology of IBD and excessive oxygen radicals are the critical factors to ulcerative inflammation tissue damage and the formation of colitis [6]. With the production of MDA, excessive NO produces oxygen free radicals that cause oxidative damage, which results in the infiltration of massive inflammatory cells

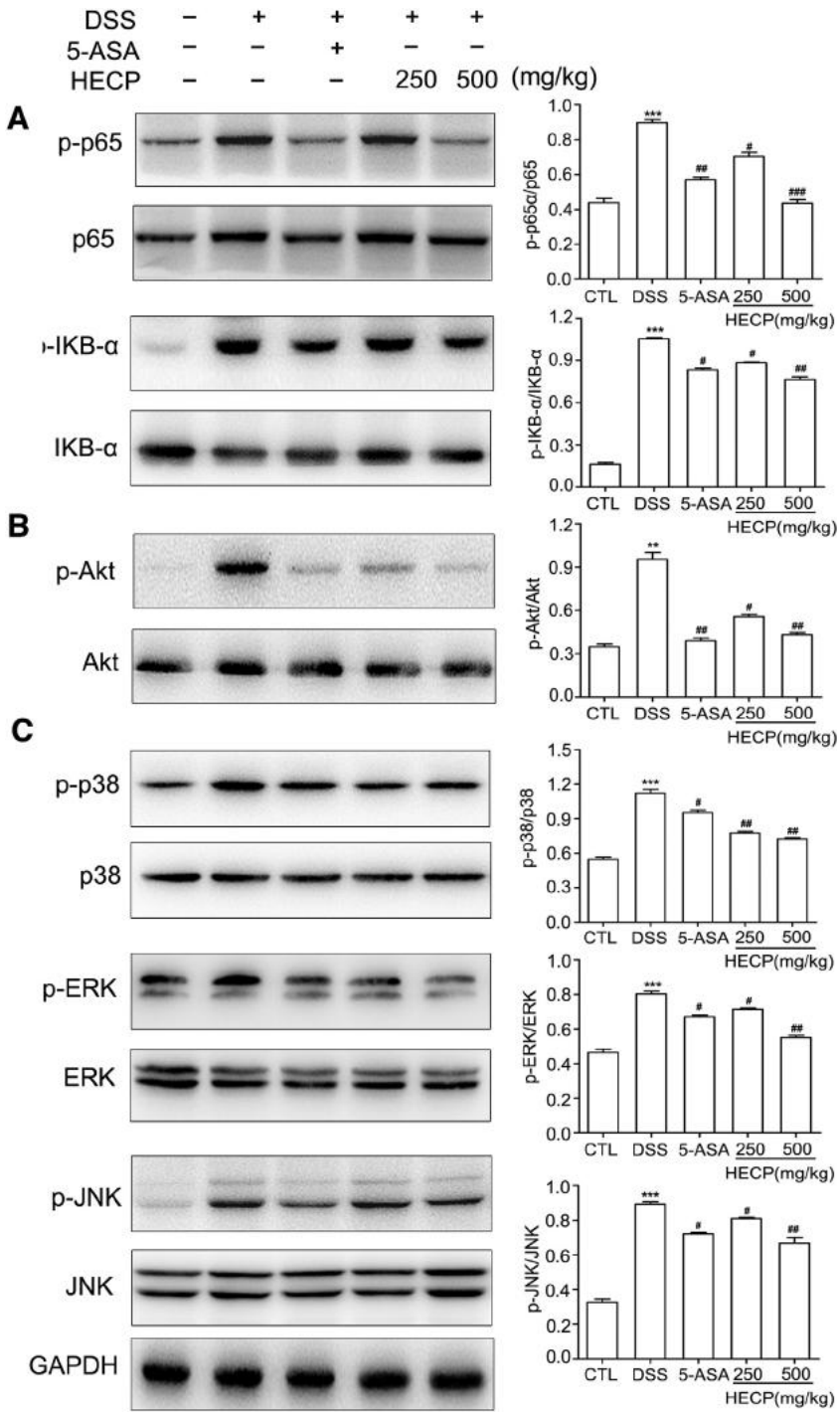

Fig. 4. Effect of HECP on the signaling pathways of NF-κB, PI3K/AKT and MAPK involved in DSS-induced colitis. Expression of (A) NF-κB, (B) PI3K/AKT and(C) MAPK were detected by Western blot. $^{*}P<.05$, $^{**}P<.01$, and $^{***}P<.001$ vs. the CTL group. $^{\#}P<.05$, $^{\#\#}P<.01$ and $^{\#\#\#}P<.001$ vs. the DSS group.

in tissue [32]. MPO, a marker of tissue injury and neutrophil infiltration, leads to mucosal disruption and ulceration [33]. This fact is documented in previous studies that MPO increased in DSS-induced colitis and has a close relationship with IBD [32,33]. We found that with HECP administration, NO and MDA contents were significantly decreased, while the T-SOD activity had a corresponding increase, showing that HECP has the ability to scavenge free radicals and increasing the anti-oxidative capacity of cells in a degree. The antioxidant enzymes (SOD, Vitamin C, coenzyme Q10, etc.) could also come from gut microbes. HECP significantly inhibited MPO activity to decrease the extensive neutrophil infiltration into the colonic tissues and repair the formation of oxidative damage in colitis. HECP exerted anti-inflammatory action by targeting major inflammatory pathways including NF-κB, MAPK, and PI3K/AKT pathways [34,35]. Activated NF-κB signaling pathway can regulate the expression of genes encoding cytokines and inflammatory enzymes such as iNOS and COX-2 [33,36]. The up-regulation of inflammatory proteins COX-2 and iNOS are drawn into the inflammatory milieu to affect colon mucosa integrity and contribute to the development of intestinal damage in colitis [37]. Tissue injury can be prevented by

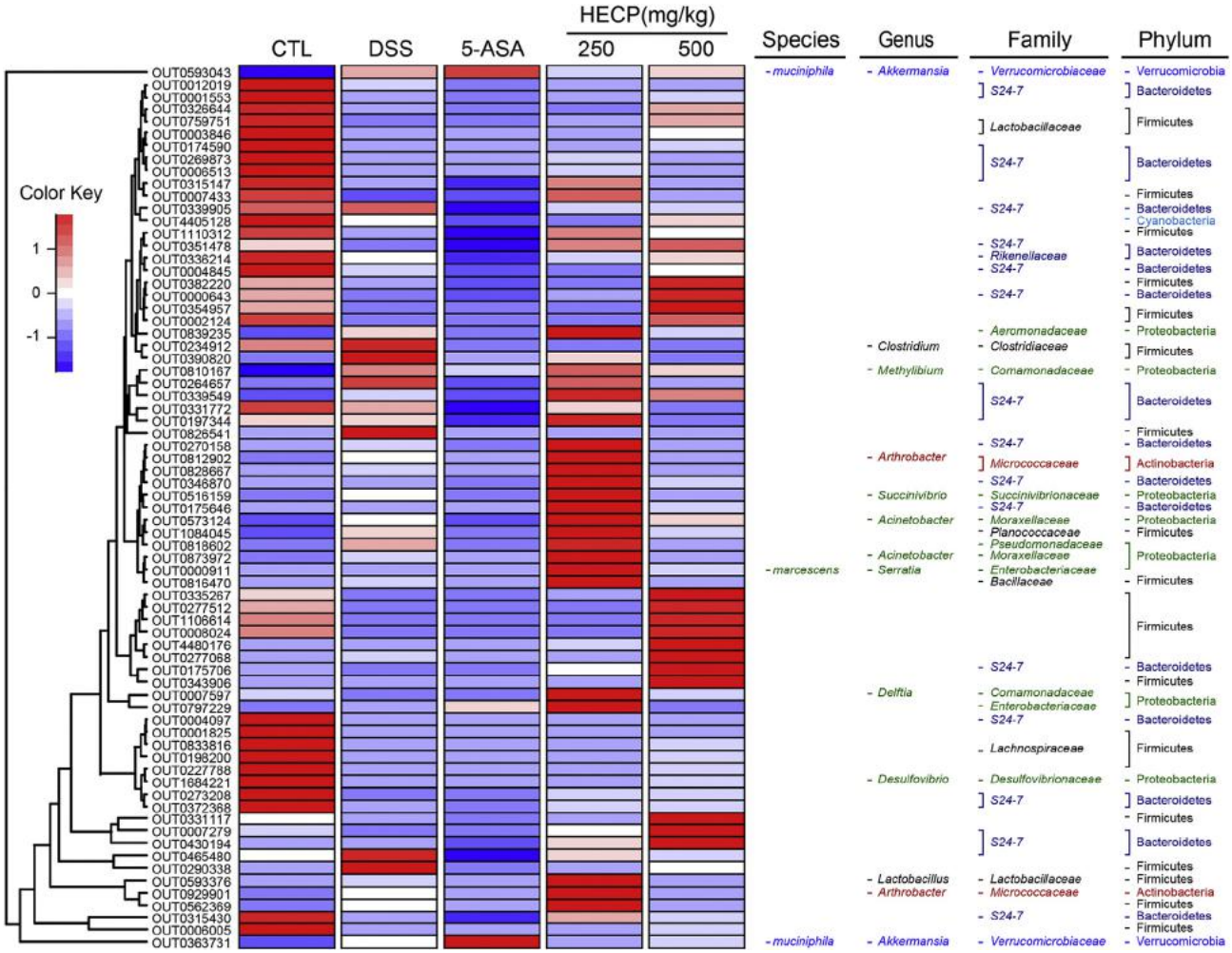

Fig. 5. Heatmap showing the abundance of 71 OTUs altered by HECP in DSS-fed mice based on RDA. The color of the spot corresponds to the normalized and log-transformed relative abundance of the OTUs. The OTUs are organized according to their phylogenetic positions. The taxonomic assignment of each OTU was determined by RDP classifier and the genus name was labeled in the graph. OTU taxonomy is shown on the left. Represented bacterial taxa information (species, genus, family and phylum) of 71 OTUs are shown on the left.

inhibiting the inducible enzymes COX-2 and iNOS that are mainly synthesized by inflammatory cells [38]. More and more evidences suggest that high levels of pro-inflammatory cytokines regulated by inflammation-related signaling pathways directly influence intestinal epithelial tissues and the development of IBD [39,40]. In addition, the inhibitory effect on NF-κB signaling pathway can be mediated through PI3K/Akt signaling that activated via Akt phosphorylation [41]. Our data indicated that HECP regulated the development of inflammation through inhibition of COX-2, iNOS, and phosphorylation of NF-κB, MAPK and PI3K/AKT in DSS treated colon tissues. Immune factors are major pathogenesis of IBD, including anti-inflammatories (IL-1, IL-10, IL-11, and transforming growth factor β) and pro-inflammatory factors (TNF-α, IL-6, IL-13, IL-17, IL-18, and IL-21) [42]. These cytokines have a wide range of functions, such as regulating immune response, stimulating hematopoiesis, participating in the differentiation and development of immune cells and tissue repair, interfacing inflammation [43]. Polysaccharides, isolated from plants, especially mushrooms, reduce and limit the damage caused by IBD through anti-inflammatory and immunosuppressive activities [13,25]. For instance, the polysaccharides of *Flammulina Velutipes* could increase the proliferation and phagocytic activity of macrophage [44], *Lentinus edodes* β-glucan inhibited the expression of pro-inflammatory factors and suppressed the phosphorylation of Elk-1 at Ser84 as well as the phosphorylation of PPARγ at Ser112 [45]. Our data showed that HECP regulated IL-1β, IL-6, TNF-α expressions and NO production in vivo and in vitro (Fig. S5). However, the detailed molecular mechanism and characteristics need to be revealed in further studies.

Intestinal inflammation may in part be a result of the intestinal bacteria, but inflammation also effects the environment and alters the bacteria. After enteritis, the intestinal barrier function is impaired and the intestinal environment changes. Subsequently, intestinal flora of dynamic balance was broken, and the characteristic of gut microbiota was changed, such as the bacterial phyla Bacteroidetes (Prevotellaceae) and TM7. Studies have shown that reducing probiotics such as *Prevotella* could make the intestinal environment even worse. Prevetellaceae might disrupt the mucosal barrier function through production of sulfatases that actively degrade mucus oligosaccharides [46,47]. Patients with IBD showed a significant reduction in the diversity of microflora [48], which is consistent with our results. Recent studies have found that colonic and systemic health improved a lot and inflammation ameliorated significantly after giving prebiotics and probiotics [13]. Mammalian intestinal bacterial communities have large variations between individual subjects but their ability to resist external disturbances is basically the same [49]. This may be related to the metabolic function of microorganisms. To an even greater extent, most humans share a core set of microbiome-encoded genes and metabolic functions [50–52]. For instance, both *Eubacterium* and *Bifidobacterium* can use germinated barley foodstuff to generate butyrate,

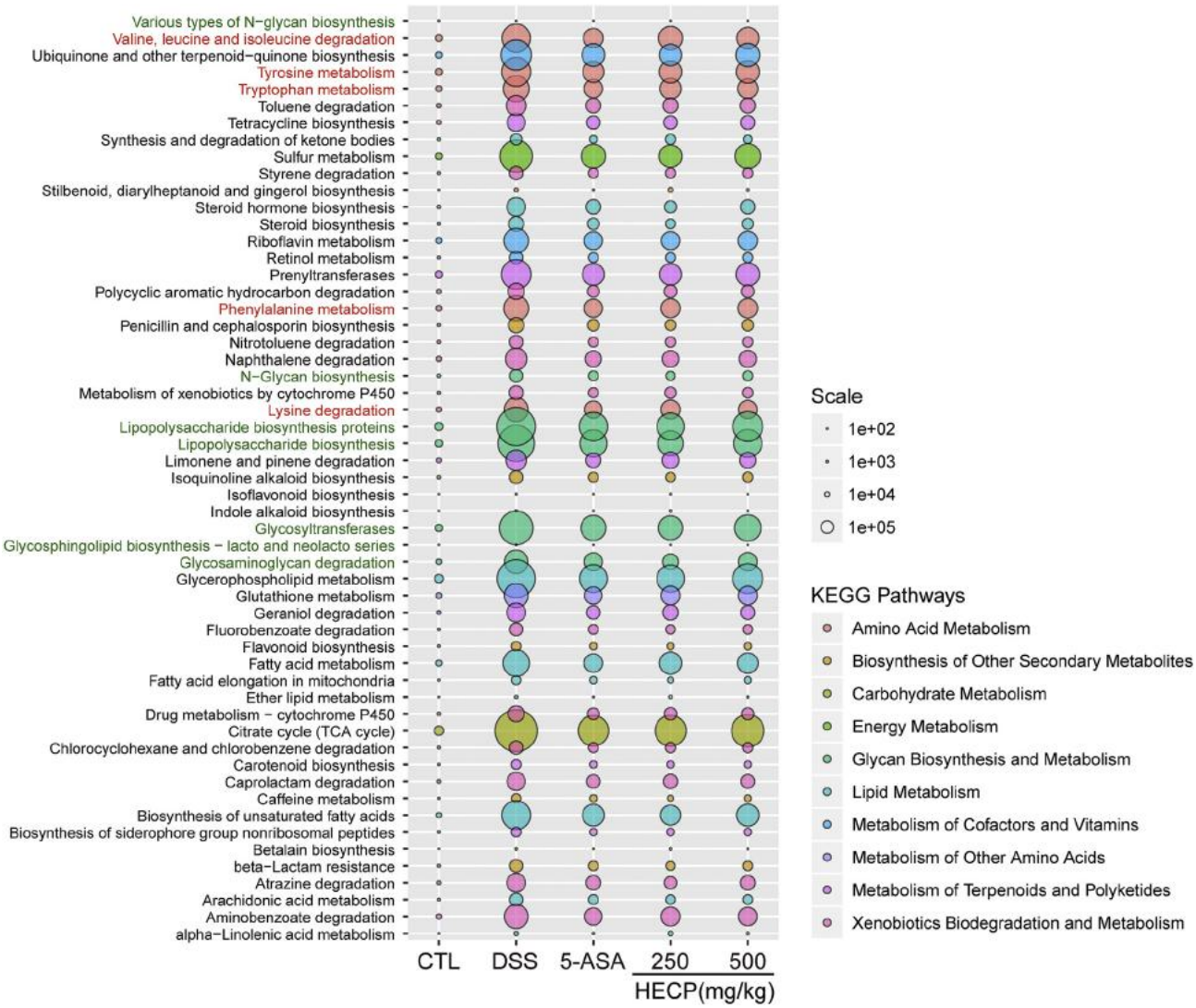

Fig. 6. PICRUSt predicted functions of KEGG categories represented in mice after HECP treatment. The diameter of bubble correlates to the read number of different KEGG pathway.

which can provide anti-inflammatory effects [53]. Our results indicated that the intestinal microbial flora of HECP 250 and 500 group were not the same, but by PICRUSt analysis, their functional genes seem to be similar. Note that the tools like PICRUSt have important limitations since strains even in the same species could have the different function. Considered about the component of HECP, glycan biosynthesis and metabolism and amino acid metabolism pathways may have major physiological meaning. Most of the polysaccharides cannot be degraded by enzymes encoded in the human genome. Microbial fermentation, however, transforms these indigestible glycans into short chain fatty acids (SCFAs), which act as nutrients for colonocytes and other gut epithelial cells [54]. A recent study showed that highly soluble glycans such as inulin may be metabolized faster by bacteria and could be processed in proximal regions of the gut, while insoluble or complicated glycans may take longer to degrade and thus reach more distal regions, which is more helpful to the colon [54,55]. Amino acids not only support the growth and survival of bacteria in the gastrointestinal tract, but also regulate the balance of energy and protein in the organism [56]. The colonic bacteria preferred amino acid substrates (lysine, arginine, glycine, valine, and isoleucine), resulting in the generation of a complex mixture of metabolic end products (others ammonia, SCFAs, branched-chain fatty acids) [56,57]. These bacterial metabolites have been shown to affect epithelial physiology by influencing the signaling pathways in epithelial cells and by modulating the host's mucosal immune system, while others also modulate the gene expression of bacteria, resulting in the production of enzymes involved in amino acid metabolism [58–60]. Additionally, in accordance with previous research, Bacteroidetes, Firmicutes, and Verrucomicrobia were the dominant phyla found in the gut (Fig. 5A), which is inconsistent with the other studies [13,61]. We speculate that the possible cause behind this finding is a difference in feeding or birth environment. In the intestine, mucins are secreted by intestinal goblet cells and build a barrier that prevent large particles, including most bacteria, from directly contacting the epithelial cell layer [62]. *Akkermansia muciniphila*, one of the mucin-degrading specialists in the gut [63], was more prevalent in mice with colitis induced by DSS [64,65]. After *Akkermansia muciniphila* overgrowth, intestinal mucins are degraded, and the intestinal epithelial cells directly face the microbes. Prebiotics, such as polysaccharides and cellulose could inhibit the excessive proliferation of *Akkermansia muciniphila*, promote the proliferation of beneficial bacteria. Gut bacteria can ferment non-digestible

polysaccharides to be SCFAs. SCFAs serve as a vital fuel source for the colonic epithelium and key regulators of immune homeostasis, especially butyrate [66,67]. In our study, we found that HECP had the same effect. In addition, HECP also altered the composition of gut microbiota by increasing the proportion of Firmicutes. Firmicutes is the main category of intestinal microorganism, and its proportion of intestinal bacteria is increased in many diseases such as obesity and cirrhosis. *Faecalibacterium prausnitzii* could secrete metabolites to exhibits anti-inflammatory effects on cellular and TNBS colitis models by blocking NF-κB B activation and IL-8 production [68].

In summary, these results indicated that HECP could ameliorate by not only adjusting the production of NO, MDA, T-SOD and MPO to repair the oxidative damage of mucosal barrier, but also down-regulating expression of COX-2, iNOS, and cytokines via blocking NF-κB, MAPK and PI3K/AKT signaling pathways in colon tissue. HECP also modulated the distribution of the intestinal flora and restored the relative abundances of vital bacteria including Clostridiales, *Akkermansia* as well as *Desulfovibri*o, and prevented the imbalance of gut microbiota in DSS-induced colitis. HECP can be used as a nutritional supplement to ameliorate colitis, with little side effects. In this study, we found that HECP might be a multi-targeted polysaccharide for protecting against IBD. On the one hand, it is potentially anti-oxidant itself to improve inflammation and excessive immune responses, on the other hand, intestinal bacteria possess distinct polysaccharide preferences and that could modulate the composition of the gut microbiota to protect the intestinal barrier integrity.

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Conflict of interest

The authors have declared no conflicts of interest.

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